

Urban coupled WRF-CMAQ (UACM)

A new version of the WRF-CMAQ combined meteorology and air quality model has been developed to include the effects of building morphology on meteorology and air quality at high resolution ($D_x \sim < 1$ km) in urban areas. The Urban parameterization is built into the PX LSM and the ACM2 PBL scheme in WRFv4.7 and the ACM2 and M3Dry dry deposition components of CMAQv5.5+. The urban model, called UACM, can be run as WRF only or coupled WRF-CMAQ.

The UACM has been developed over the past 6 years in three stages. The first stage was the development of building effects on winds in a modified version of the ACM2 PBL model in WRF (Dy et al 2019). The second stage involved the implementation of the thermal effects of the building canopy including the interaction of SW and LW radiation with the walls of street canyons and surface heat fluxes from roof, walls, and ground (Bhautmage et al., 2022) which is mostly based on the work of Mason 2000 and Martilli et al 2002. These first two stages idealized the urban building morphology as parallel infinite street canyons described by building height H , building width B , and street width W . The model required inputs for average building height, frontal area density (λ_f), and the plan area density (λ_p) from which B and W are computed as described by Dy et al (2019). The third stage of development included implementation of a second street canyon direction so that the building morphology is modeling as a grid intersecting streets which define blocks with average building height in each grid cell and the length and width of blocks proscribed by the street grid. Note that this idealized morphology is most appropriate for dense urban centers of large cities such as NYC and Chicago in the US which are characterized by deep street canyons. The third stage of development also included modification to the CMAQ model to create a consistent on-line coupled meteorology-air quality urban model. Modifications to CMAQ included development of dry deposition to wall and roofs of buildings based on momentum and heat fluxes to walls and roofs computed in the urban WRF.

There are two ways to define the building characteristics in UACM. The more accurate method is to acquire detailed building characteristics from urban planning datasets. Using GIS processing to derive average building height and building plan area for each grid cell. Then use GIS to process detailed street data to compile the direction and length of every street in each grid cell. A simple Fortran code is included to process street data to compute the top two most prevalent street directions (within 5 degree sectors) and the block widths associated with the street directions. Python programs are included that add the building data and the street data to the wrfinput files.

The second method uses WUDAPT LCZ categories to provide the building and street parameters from lookup table in the code. To use this option the 11 LCZ categories are added to the geogrid files. Scripts are included in LCZ_proc that create NLCD51 from NLCD40 and MODIS61. Note that the LCZ categories are independent from the other landuse categories. Therefore, the standard LU data sets that include LCZ data such as MODIS 61 should not be used directly.

Changes to WRF include

- Major changes to module_sf_pxlsm.F and module_bl_acm.F
- Minor changes in module_surface_driver.F, module_pbl_driver.F, module_sf_pxlsm_data.F, module_radiation_driver.F, module_physics_init.F
- In dyn_em:

- Solve_em.F has been modified to allow for coupled wrf-cmaq to run on the nested domain but wrf only on the mother domain. This does not affect WRF only runs
- Module_first_rk_step_part1.F is also modified
- Registry.EM.COMMON

The UACM is invoked by setting the namelist flag in &physics ifuacm = 1 for which ever domain you want.

```
&wrf_cmaq
wrf_cmaq_option      = 2,
wrf_cmaq_freq        = 5,
met_file_tstep       = 10000,
direct_sw_feedback   = .false.,
feedback_restart     = .false.,
/
```

wrf_cmaq_option = 0 run WRF only
 = 1 run WRF only and produce GRID and MET files as well
 = 2 run WRF-CMAQ coupled model w/o producing GRID and MET files
 = 3 run WRF-CMAQ coupled model w/ producing GRID and MET files

Changes to CMAQ include:

```
vdifacmx.f
twoway_aqprep.F90
subhfile.F
m3dry.F
eddyx.F
DEPV_DEFN.F
coupler_mod.F90
ASX_DATA_MOD.F
aero_subs.F
aero_depv.F
```